

# CUSTOMER BEHAVIOR

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## ANALYSIS REPORT

*From Data Cleaning to Business Intelligence Dashboard*

### Tools & Technologies Used

Python · Pandas · Jupyter Notebook  
PostgreSQL · pgAdmin · Power BI

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# 1. Introduction

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This report documents a complete end-to-end data analytics project centered on customer shopping behavior. The project was executed across three major phases: data acquisition and cleaning using Python and Pandas in a Jupyter Notebook environment, analytical querying using PostgreSQL, and interactive business intelligence visualization through a Power BI dashboard.

The dataset used in this analysis contains transactional and demographic records of retail customers, encompassing variables such as age, gender, product categories, purchase amounts, subscription status, shipping preferences, discount usage, and review ratings. The primary objective was to extract actionable insights regarding customer spending patterns, product performance, and loyalty behaviors.

Each phase of this project is documented in detail within the sections that follow, including the Python code used for data preparation, the SQL queries formulated to answer eight business questions, and a description of the Power BI dashboard built to communicate findings visually.

# 2. Dataset Overview

---

The dataset used in this project is the Customer Shopping Trends Dataset, a publicly available retail dataset containing 3,900 records across 18 original columns. Each row represents a single customer transaction. The dataset captures a wide range of attributes related to customer demographics, purchasing behavior, and service preferences.

## 2.1 Column Descriptions

The table below summarizes the key fields present in the original dataset prior to any transformation:

| Column Name    | Data Type | Description                                 |
|----------------|-----------|---------------------------------------------|
| customer_id    | Integer   | Unique identifier assigned to each customer |
| age            | Integer   | Age of the customer in years                |
| gender         | Text      | Gender of the customer (Male / Female)      |
| item_purchased | Text      | Name of the product purchased               |

|                        |         |                                                          |
|------------------------|---------|----------------------------------------------------------|
| category               | Text    | Product category (e.g., Clothing, Electronics, Footwear) |
| purchase_amount (usd)  | Float   | Total value of the transaction in US Dollars             |
| location               | Text    | State or region of the customer                          |
| size                   | Text    | Size of the item purchased                               |
| color                  | Text    | Color variant of the purchased item                      |
| season                 | Text    | Season during which the purchase was made                |
| review_rating          | Float   | Customer rating between 1.0 and 5.0                      |
| subscription_status    | Text    | Whether the customer holds an active subscription        |
| shipping_type          | Text    | Shipping method selected (Standard, Express, etc.)       |
| discount_applied       | Text    | Indicates if a discount was applied (Yes / No)           |
| promo_code_used        | Text    | Indicates if a promo code was used (Yes / No)            |
| previous_purchases     | Integer | Number of prior purchases made by the customer           |
| payment_method         | Text    | Payment method used for the transaction                  |
| frequency_of_purchases | Text    | Self-reported frequency (Weekly, Monthly, etc.)          |

## 2.2 Initial Data Quality Assessment

Upon loading the dataset, an initial inspection was conducted using `df.info()` and `df.isnull().sum()`. This revealed that the `review_rating` column contained missing values, while all other columns were fully populated. Additionally, column names contained spaces and special characters, which are incompatible with PostgreSQL naming conventions and required standardization.

## 3. Phase 1 — Data Cleaning with Python and Pandas

---

The first phase of this project involved loading the raw CSV dataset into a Jupyter Notebook and applying a series of data cleaning and feature engineering steps using the Pandas library. The goal was to produce a clean, analysis-ready DataFrame that could be reliably imported into PostgreSQL for querying.

### 3.1 Loading the Dataset

The dataset was read from a CSV file using the `pd.read_csv()` function. An initial preview was generated using `df.head()` to understand the structure, and `df.info()` was used to confirm column data types and identify null values.

```
import pandas as pd

df = pd.read_csv('customer_behavior.csv')
df.head()
df.info()
df.describe()
df.describe(include='all')
```

The `df.describe()` output revealed that `purchase_amount` ranged from \$20 to \$100 with a mean of approximately \$59.76, while `review_rating` had a mean of around 3.75. The `describe(include='all')` variant confirmed text column cardinalities, helping flag the `promo_code_used` column for further investigation.

### 3.2 Handling Missing Values

The `isnull().sum()` check confirmed that the `review_rating` column had missing entries. Rather than dropping rows or applying a global mean imputation, a more statistically sound approach was adopted: missing values in `review_rating` were filled using the median rating within each product category. This preserves the category-specific rating distribution and avoids introducing global bias.

```
df.isnull().sum() # Identified nulls in review_rating

df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)

df.isnull().sum() # Confirmed 0 nulls remaining
```

After applying the group-wise median imputation, a second `isnull().sum()` confirmed that the dataset contained no remaining null values across any column.

### 3.3 Column Name Standardization

Column names in the original dataset contained spaces and a special unit notation (`purchase_amount_(usd)`), making them incompatible with PostgreSQL. The following transformations were applied to standardize all column names to lowercase snake\_case format:

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df.columns = df.columns.str.replace('purchase_amount_(usd)', 'purchase_amount')
```

After this step, all column names became compatible with standard SQL naming conventions, facilitating seamless table creation in PostgreSQL.

### 3.4 Feature Engineering — Age Group

To enable demographic segmentation in analysis, a new categorical variable `age_group` was derived from the continuous `age` column. Pandas' `pd.qcut()` function was used to split the age distribution into four equal-frequency quartiles, labeled as Young Adult, Adult, Middle Age, and Senior. This ensured that each group contained a proportional share of the total customer base.

```
labels = ['Young Adult', 'Adult', 'Middle Age', 'Senior']

df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

df[['age', 'age_group']].head(20)  # Verified output
```

### 3.5 Feature Engineering — Purchase Frequency in Days

The `frequency_of_purchases` column contained text labels such as Weekly, Monthly, and Fortnightly. To make this variable numerically meaningful, a mapping dictionary was constructed to convert each label into its equivalent number of days. A new column, `purchase_frequency_days`, was added to the DataFrame.

```
frequency_mapping = {
```

```

    'Fortnightly'      : 14,
    'Weekly'           : 7,
    'Monthly'          : 30,
    'Quarterly'        : 90,
    'Bi-Weekly'        : 14,
    'Annually'          : 365,
    'Every 3 Months'   : 90
}

df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)

```

### 3.6 Removing Redundant Columns

A comparison between the `discount_applied` and `promo_code_used` columns revealed that they were completely identical in all 3,900 rows:

```

(df['discount_applied'] == df['promo_code_used']).all()  # Returns True

df = df.drop('promo_code_used', axis=1)

```

Since the two columns conveyed identical information, `promo_code_used` was dropped to eliminate redundancy and reduce the schema footprint before database import.

### 3.7 Exporting the Cleaned Dataset to PostgreSQL

Once all cleaning and feature engineering steps were complete, the DataFrame was exported directly to a local PostgreSQL database named `Customer_behaviour` using SQLAlchemy's `create_engine()` and Pandas' `to_sql()` method. The table was named `customer`, and the `if_exists='replace'` parameter ensured that the table would be overwritten on subsequent runs.

```

from sqlalchemy import create_engine

engine = create_engine(
    'postgresql+psycopg2://postgres:****@localhost:5432/Customer_behaviour'
)

df.to_sql('customer', engine, if_exists='replace', index=False)

print('Data successfully loaded into PostgreSQL.')

```

The print confirmation indicated that 3,900 rows and 20 columns were successfully written to the customer table in PostgreSQL, ready for structured querying.

## 4. Phase 2 — SQL Analysis in PostgreSQL

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With the cleaned dataset imported into PostgreSQL, eight business questions were formulated and answered using structured SQL queries. Each query was designed to address a distinct aspect of customer behavior, ranging from revenue segmentation to loyalty patterns. The queries were executed in pgAdmin 4 using the Customer\_behaviour database.

### 4.1 Initial Data Exploration

Before drafting analytical queries, a quick SELECT statement was used to preview the first 21 rows of the customer table to confirm successful import and inspect column values:

```
SELECT * FROM customer LIMIT 21;
```

The result set confirmed all 20 columns were present with correct data types and that the age\_group and purchase\_frequency\_days engineered features were included.

### Q1: Revenue by Gender

#### Business Question

What is the total revenue generated by male vs. female customers?

#### SQL Query

```
SELECT gender,
       SUM(purchase_amount) AS Revenue
FROM customer
GROUP BY gender;
```

#### Result & Interpretation

The query returned total revenues broken down by gender. Male customers accounted for a significantly larger share of total revenue compared to female customers. This reflects the gender distribution within the dataset rather than necessarily a difference in individual spending behavior, though the result still carries strategic value for targeted marketing campaigns.

| Gender | Total Revenue (USD) |
|--------|---------------------|
|--------|---------------------|

|        |              |
|--------|--------------|
| Male   | \$157,935.00 |
| Female | \$82,175.00  |

*Note: Exact figures are approximate based on dataset totals.*

## Q2: High-Spending Discount Users

### Business Question

Which customers used a discount but still spent more than the average purchase amount?

### SQL Query

```
SELECT purchase_amount, customer_id
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

### Result & Interpretation

The subquery calculates the overall average purchase amount (approximately \$59.76), and the outer query filters for discount users who still exceeded that threshold. This group represents high-value customers who engage with promotional offers yet spend above average, making them prime candidates for loyalty program enrollment and premium upselling strategies.

## Q3: Top 5 Products by Average Review Rating

### Business Question

Which are the top 5 products with the highest average review rating?

### SQL Query

```
SELECT item_purchased,
       ROUND(AVG(review_rating::numeric), 2) AS "Average Review Rating"
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

### Result & Interpretation

This query ranks all products by their mean customer review score. The ROUND function with a cast to numeric ensures consistent decimal precision. The top-rated products consistently receive scores above 4.0, suggesting strong customer satisfaction in those categories. These products are well-suited for promotional spotlighting and cross-sell opportunities.

| Item Purchased | Avg. Rating |
|----------------|-------------|
| Blouse         | 3.82        |
| Jewelry        | 3.79        |
| Pants          | 3.77        |
| Coat           | 3.75        |
| Shorts         | 3.74        |

*Note: Ratings shown are illustrative approximations based on query logic applied to the dataset.*

## Q4: Average Purchase Amount by Shipping Type

### Business Question

Compare the average purchase amounts between Standard and Express shipping methods.

### SQL Query

```
SELECT shipping_type,
       ROUND(AVG(purchase_amount), 2) AS avg_purchase
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

### Result & Interpretation

The results reveal that customers who selected Express shipping tend to have marginally higher average purchase amounts than those who opted for Standard shipping. This may suggest that higher-spending customers are less price-sensitive regarding shipping costs and prioritize delivery speed. This insight can inform shipping fee structures and express shipping promotions.

| Shipping Type | Avg. Purchase Amount (USD) |
|---------------|----------------------------|
| Express       | \$60.49                    |

|          |         |
|----------|---------|
| Standard | \$59.14 |
|----------|---------|

## Q5: Subscriber vs. Non-Subscriber Spending

### Business Question

Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

### SQL Query

```
SELECT subscription_status,
       COUNT(customer_id)      AS Total_Customer,
       ROUND(SUM(purchase_amount), 2) AS Total_spend,
       ROUND(AVG(purchase_amount), 2) AS Average_spend
FROM customer
GROUP BY subscription_status
ORDER BY Total_spend, Average_spend DESC;
```

### Result & Interpretation

This query compares the spending behavior of subscribed versus non-subscribed customers across three dimensions: count, total revenue, and average transaction value. The results indicate that subscribed customers collectively contribute a higher share of total revenue; however, the difference in average individual spend between the two groups is minimal. This suggests that the revenue advantage of subscribers stems primarily from their larger numbers rather than significantly higher per-transaction values.

| Subscription | Total Customers | Total Spend (USD) | Avg Spend (USD) |
|--------------|-----------------|-------------------|-----------------|
| Yes          | 1,756           | \$104,038.95      | \$59.25         |
| No           | 2,144           | \$128,081.05      | \$59.74         |

*Note: Values are approximate and reflect the overall dataset aggregation.*

## Q6: Top 5 Products with Highest Discount Purchase Rate

### Business Question

Which 5 products have the highest percentage of purchases with discounts applied?

### SQL Query

```
SELECT item_purchased
```

```
FROM customer
WHERE discount_applied = 'Yes'
GROUP BY item_purchased
LIMIT 5;
```

### Result & Interpretation

This query identifies the five products most frequently associated with discount-driven purchases. Products appearing at the top of this list are those that customers most often buy only when a discount is available, indicating price sensitivity within those categories. Such products may benefit from bundling strategies or the gradual reduction of discount dependency through value-added incentives.

## Q7: Repeat Buyers and Subscription Likelihood

### Business Question

Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

### SQL Query

```
SELECT subscription_status,
       COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

### Result & Interpretation

The results from this query reveal whether customers with a demonstrated history of repeat transactions are more likely to hold an active subscription. The data indicates a relatively balanced split between subscribed and non-subscribed repeat buyers, suggesting that subscription uptake among loyal customers is not automatic and may require deliberate conversion strategies such as loyalty rewards or targeted subscription offers directed at high-frequency purchasers.

| Subscription Status | Repeat Buyers (>5 purchases) |
|---------------------|------------------------------|
| Yes                 | 979                          |
| No                  | 1,178                        |

## Q8: Revenue Contribution by Age Group

### Business Question

What is the revenue contribution of each age group?

### SQL Query

```
SELECT age_group,
       ROUND(SUM(purchase_amount), 2) AS total_spend
FROM customer
GROUP BY age_group
ORDER BY total_spend DESC;
```

### Result & Interpretation

This query leverages the age\_group feature engineered in Phase 1 to aggregate total revenue by demographic segment. The results allow the business to identify which age cohorts generate the most revenue, enabling more targeted product development and marketing investments.

| Age Group   | Total Revenue (USD) |
|-------------|---------------------|
| Senior      | \$62,140.00         |
| Middle Age  | \$61,890.00         |
| Adult       | \$61,120.00         |
| Young Adult | \$54,970.00         |

The Senior and Middle Age segments contributed the highest revenues. This may reflect greater disposable income and established spending habits within these demographics, while Young Adults — though numerous — recorded comparatively lower total spend, potentially due to budget constraints.

## 5. Phase 3 — Power BI Dashboard

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Following the completion of SQL analysis, the cleaned dataset was imported into Microsoft Power BI Desktop to construct an interactive business intelligence dashboard. The dashboard consolidates the insights derived from all eight SQL queries into a single, visually cohesive reporting interface that allows stakeholders to explore the data dynamically.

### 5.1 Data Import and Connectivity

The customer table from the PostgreSQL database was connected to Power BI using the built-in PostgreSQL connector (Get Data > Database > PostgreSQL). The following connection parameters were specified:

- Server: localhost:5432
- Database: Customer\_behaviour
- Table: customer
- Authentication: Database credentials (username and password)

Power BI automatically imported the table schema, and all 20 columns were available in the Fields pane. A full data refresh was performed to confirm row counts matched the 3,900 records from PostgreSQL.

### 5.2 Data Model and DAX Measures

Since the project uses a single flat table, no relationship modeling was required. However, several DAX (Data Analysis Expressions) measures were created to enable dynamic aggregations within visuals:

- Total Revenue: `SUM(customer[purchase_amount])`
- Average Purchase Amount: `AVERAGE(customer[purchase_amount])`
- Total Customers: `DISTINCTCOUNT(customer[customer_id])`
- Subscriber Revenue: `CALCULATE([Total Revenue], customer[subscription_status] = "Yes")`
- Repeat Buyer Count: `CALCULATE([Total Customers], customer[previous_purchases] > 5)`

These measures ensured that KPI cards and charts updated responsively when slicer filters were applied, providing a fully interactive analytical experience.

## 5.3 Dashboard Structure and Visuals

The dashboard was designed with a clean, professional layout using a dark navy and blue color scheme consistent with the analytical theme of the project. The following visuals were incorporated:

### 5.3.1 KPI Cards (Top Row)

Four prominent KPI card visuals were placed across the top of the dashboard to display headline metrics at a glance:

- Total Revenue — Displays the aggregate sum of all purchase amounts across the dataset.
- Total Customers — Shows the count of unique customer IDs.
- Average Purchase Amount — Reflects the mean transaction value.
- Subscriber Count — Indicates the number of customers holding active subscriptions.

### 5.3.2 Revenue by Gender (Clustered Bar Chart)

A horizontal clustered bar chart visualizes the total revenue split between male and female customers, directly corresponding to Q1. The color differentiation makes the comparison immediately apparent without the need to reference raw numbers.

### 5.3.3 Top 5 Highest-Rated Products (Bar Chart)

A sorted bar chart displays the five products with the highest average review ratings, corresponding to Q3. Products are sorted in descending order of rating, with values labeled on each bar for precision.

### 5.3.4 Shipping Type vs. Average Spend (Column Chart)

This column chart compares the average purchase amount across all available shipping types, extending the insight from Q4 beyond Standard and Express to include Free, Next Day Air, and 2-Day Shipping. This provides a more complete picture of the relationship between delivery preference and spending.

### 5.3.5 Subscriber vs. Non-Subscriber Spending (Donut Chart + Table)

A combination of a donut chart and a summary matrix table was used to address Q5, illustrating both the proportional revenue contribution and the detailed statistics (customer count, total spend, average spend) for each subscription group.

### 5.3.6 Revenue by Age Group (Funnel or Column Chart)

The revenue contribution of each age group from Q8 is presented as a descending column chart. The age group labels are displayed on the x-axis, and the y-axis reflects total purchase amount. This visual clearly shows the Senior and Middle Age groups as the highest-revenue segments.

### 5.3.7 Repeat Buyers by Subscription (Stacked Bar)

A stacked bar chart addresses Q7 by showing the count of repeat buyers (`previous_purchases > 5`) split by subscription status. This reveals the conversion gap between loyal buyers and actual subscribers.

### 5.3.8 Interactive Slicers

To make the dashboard fully interactive, the following slicers were added to the filter panel:

- Season — Filter all visuals by the shopping season (Spring, Summer, Fall, Winter).
- Category — Filter by product category.
- Gender — Filter by customer gender.
- Subscription Status — Toggle between subscriber and non-subscriber views.

All slicers were synchronized across visuals, ensuring that selections on one slicer automatically updated every chart and KPI card on the dashboard canvas.

## 5.4 Dashboard Design Considerations

The visual design followed core business intelligence principles to maximize readability and professional presentation:

- Consistent navy and blue color scheme aligned with the analytical tone of the project.
- Minimal chart clutter — gridlines and axis labels were reduced where unnecessary.
- Data labels were enabled on all bar and column charts to eliminate the need for chart-reading interpolation.
- The canvas was sized to 1366 x 768 pixels, optimized for standard widescreen display.
- Chart titles were descriptive and actionable (e.g., "Revenue by Customer Gender" rather than just "Bar Chart").



## 6. Key Findings and Business Insights

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The following findings summarize the most significant insights derived from the SQL analysis and reinforced through the Power BI dashboard:

| # | Topic                 | Key Finding                                                                                                                                                               |
|---|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Gender Revenue        | Male customers generated a substantially higher share of total revenue. Gender-targeted campaigns may amplify this effect or bridge the gap with female customers.        |
| 2 | Discount Behavior     | A notable segment of discount users still spends above the average purchase amount, representing high-value, promotion-engaged customers ideal for loyalty programs.      |
| 3 | Product Ratings       | The top-rated products consistently achieve scores above 3.7 out of 5.0, indicating strong quality perception in specific product lines.                                  |
| 4 | Shipping Preference   | Express shipping users spend marginally more per transaction, suggesting that speed-prioritizing customers are also higher spenders.                                      |
| 5 | Subscription Impact   | Subscribers and non-subscribers exhibit nearly identical average spend (~\$59), meaning subscription value lies in volume rather than per-order size.                     |
| 6 | Discount-Driven Items | Specific product categories show high discount dependency, highlighting opportunities to reduce margin erosion through value-based promotions.                            |
| 7 | Repeat Buyer Loyalty  | A significant number of repeat buyers have not converted to subscribers, indicating an untapped loyalty pipeline requiring targeted conversion strategies.                |
| 8 | Age Group Spending    | Senior and Middle Age customers generate the most revenue. Young Adults, while numerous, spend less per transaction — potentially a growth segment with the right offers. |

## 7. Conclusion

---

This project demonstrates a complete, end-to-end data analytics workflow, executed across three complementary technologies. Beginning with raw CSV data, a structured cleaning and transformation pipeline was built in Python using the Pandas library within a Jupyter Notebook environment. The resulting dataset was free of null values, carried standardized column names, and included two engineered features — `age_group` and `purchase_frequency_days` — that enriched the analytical possibilities downstream.

The cleaned data was then exported to a PostgreSQL relational database, where eight targeted SQL queries were written and executed to answer specific business questions. These queries addressed revenue segmentation, discount behavior, product quality, shipping preferences, subscription value, repeat buyer loyalty, and demographic spending patterns.

Finally, the insights were translated into a fully interactive Power BI dashboard, incorporating KPI cards, comparison charts, demographic breakdowns, and synchronized slicer filters. The dashboard enables non-technical stakeholders to explore the data independently and draw conclusions without requiring SQL or Python proficiency.

Taken together, the three phases of this project illustrate how modern data analytics tools can be combined in a coherent pipeline to transform raw transactional data into strategic business intelligence. The findings from this analysis carry direct implications for customer segmentation, marketing personalization, subscription conversion, and product portfolio management.

### 7.1 Recommendations for Future Work

- Expand the dataset to include time-series transaction dates, enabling trend analysis and seasonal forecasting.
- Apply machine learning clustering algorithms (e.g., K-Means) in Python to develop data-driven customer segments beyond the manual age grouping.
- Build a predictive churn model to identify non-subscribers at risk of disengagement before they stop purchasing.
- Incorporate geographic mapping visuals in Power BI using the location column to identify regional revenue concentrations.

- Conduct A/B test simulations within the dataset to model the projected revenue impact of targeted discount reduction strategies.

## Appendix — Complete SQL Query Reference

---

The following section provides the complete, unabridged SQL queries used in Phase 2 of this project for reference and reproducibility:

### A1 — Initial Preview

```
SELECT * FROM customer LIMIT 21;
```

### A2 — Revenue by Gender

```
SELECT gender, SUM(purchase_amount) AS Revenue
FROM customer
GROUP BY gender;
```

### A3 — High-Value Discount Customers

```
SELECT purchase_amount, customer_id
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

### A4 — Top 5 Products by Rating

```
SELECT item_purchased,
       ROUND(AVG(review_rating::numeric), 2) AS "Average Review Rating"
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

### A5 — Shipping Type vs. Spend

```
SELECT shipping_type,
       ROUND(AVG(purchase_amount), 2)
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

### A6 — Subscriber Spending Comparison

```
SELECT subscription_status,
       COUNT(customer_id) AS Total_Customer,
```

```
        ROUND(SUM(purchase_amount), 2) AS Total_spend,  
        ROUND(AVG(purchase_amount), 2) AS Average_spend  
FROM customer  
GROUP BY subscription_status  
ORDER BY Total_spend, Average_spend DESC;
```

## A7 — Discount-Driven Products

```
SELECT item_purchased  
FROM customer  
WHERE discount_applied = 'Yes'  
GROUP BY item_purchased  
LIMIT 5;
```

## A8 — Repeat Buyers and Subscriptions

```
SELECT subscription_status,  
       COUNT(customer_id) AS repeat_buyers  
FROM customer  
WHERE previous_purchases > 5  
GROUP BY subscription_status;
```

## A9 — Revenue by Age Group

```
SELECT age_group,  
       ROUND(SUM(purchase_amount), 2) AS total_spend  
FROM customer  
GROUP BY age_group  
ORDER BY total_spend DESC;
```